Fixed-order trilepton analysis of $t\bar{t}W^{\pm}$

Truth observables and exclusive fiducial bottom-jet categories

fNLO fixed-order template

19 October 2026

Abstract

The HwU analysis `analysis_HwU_pp_ttxw_trilepton.o` contains 581 histograms for the $2e1\mu + 3\nu + b\bar{b}$ final state. It retains inclusive truth observables and supplies 136 fiducial observables in each of the exclusive zero-, one-, and two-bottom-jet categories. The fiducial observables use charged leptons, recombined jets, and total missing transverse momentum, without fitting the neutrinos or reconstructing top and W candidates from missing momentum. All active weights are filled with the same measurement function for additive and multiplicative events and their subtraction counterevents.

1 Event content and interfaces

The analysis accepts two status-one electrons or positrons, one muon of either charge, three light neutrinos, one bottom parton and one antibottom parton, together with additional QCD radiation. The charged-lepton total charge is $q_{3\ell} = \pm 1$, with two same-sign leptons and one opposite-sign lepton. Both $e^+e^-\mu^\pm$ and $e^\pm e^\pm \mu^\mp$ patterns are supported. Identification uses PDG codes, not record position or parent ancestry. Electron and positron distributions always refer to their explicit charges; a two-electron event of one charge contributes twice to that species' single-particle distribution. Muon distributions include both charges, with additional charge-separated muon plots.

The module and its matching ABI bridge provide `analysis_begin`, `analysis_end`, `analysis_fill`, and the variable-particle-count `analysis_fill_multiplicative` entry point. The latter accommodates simultaneous production and decay radiation. Select the analysis with

```
FO_ANALYSIS_FORMAT=HwU
FO_ANALYSE=analysis_HwU_pp_ttxw_trilepton.o
```

The launcher automatically adds its matching bridge.

2 Common fiducial selection

The three exclusive categories share the following lepton selection:

- each charged lepton has $p_{\mathrm{T}} \geq 20$ GeV and $|\eta| < 2.5$, independently of its flavour and charge;
- when the electron pair is opposite sign, $m_{ee} > 12$ GeV;
- every selected lepton is separated from every accepted central jet by

$$\Delta R(\ell, j) \geq \min\left(0.4,\ 0.04 + \frac{10\,\mathrm{GeV}}{p_{\mathrm{T}}(\ell)}\right).$$

There are no Z-mass vetoes on the dilepton or three-lepton invariant mass. There is no minimum number of jets or tagged jets in the common selection, and no missing-transverse-momentum cut. In particular, an event with no accepted jets can populate the zero-bottom-jet category. The separation criterion is a specified partonic isolation proxy, not a detector simulation.

All status-one QCD partons are clustered with anti-k_T, $R = 0.4$, and four-vector recombination. Accepted jets have $p_{\mathrm{T}} \geq 25$ GeV and $|\eta| < 2.5$. A bottom jet contains the unique bottom parton and not the antibottom parton; an antibottom jet contains the unique antibottom parton and not the bottom parton. A jet containing both is excluded

from the bottom-tag count and all bottom/antibottom-specific distributions. It remains an ordinary jet for jet counts, scalar jet activity and lepton–jet separation. These definitions are appropriate to the supported fixed-order final state, which has one primary bottom and one primary antibottom; they do not define an arbitrary-multiplicity flavour algorithm for additional $b\bar{b}$ pairs.

The accepted tag count N_b is exactly zero, one, or two. Each event passing the common selection fills precisely one of the three rate histograms, `fiducial 0b rate`, `fiducial 1b rate`, or `fiducial 2b rate`. The categories are exclusive, not "at least" requirements. A species or pair observable is filled only if its required accepted objects exist; missing objects are not represented by zero momentum. Thus the zero-tag category has lepton and event observables but no tagged-jet entries. In the one-tag category only the accepted tag's species and combinations are filled.

3 Generation cuts in the current study exports

The current `PROC_fNLO_loop_sm-no_b_mass_2` and `PROC_fNLO_loop_sm-no_b_mass_3` exports also apply the common fiducial baseline in `SubProcesses/cuts.f90`. The public analysis helper `passes_fiducial_baseline` reuses the lepton acceptance, low-mass threshold and jet-isolation selection above. It imposes no bottom-tag count or jet-multiplicity requirement. Bottom jets participate in lepton isolation as ordinary accepted jets, including jets containing both bottom flavours.

The helper is called after the longitudinal boost to the lab and before the decay-product mask for generic run-card cuts. Consequently the custom baseline remains active with `cut_decays=False`. It is evaluated separately for each real event and subtraction counterevent, including variable-size multiplicative events; a failing Born projection does not veto a passing real configuration. The process makefiles make `cuts.o` depend on the analysis module. These are process-local changes: a fresh export needs the same call and dependency added to enable this generation selection.

With these generation cuts, all histograms, including those labelled `inclusive` or `truth`, are conditional on passing the common fiducial baseline. Their rates and cutflows therefore differ from runs integrated without that baseline. The three exclusive category rates still sum to the selected total. New run names must be used for this selection.

4 Fiducial observables

Each category has the same booked observable catalogue and its own `fiducial 0b`, `fiducial 1b`, or `fiducial 2b` title prefix.

Objects	Observables
Electrons, positrons, muons; also μ^+ and μ^- separately	p_{T}, pseudorapidity η, rapidity y, azimuth ϕ, energy E.
Accepted bottom and antibottom jets separately	p_{T}, η, y, ϕ, E, invariant jet mass.
Charged-lepton pairs e^-e^+, e^-e^-, e^+e^+, $e^-\mu$, $e^+\mu$	Invariant mass, pair p_{T}, pair rapidity, ΔR, $\lvert\Delta\phi\rvert$, $\lvert\Delta\eta\rvert$. Each unordered physical pair is filled once.
Same-sign pair and leading opposite-sign pair	The same six pair observables; the leading opposite-sign pair combines the unique opposite-sign lepton with the highest-p_{T} same-sign lepton.
Electron, positron, or muon with either accepted tagged-jet species	The same six pair observables, for all six species combinations.
Accepted bottom–antibottom jet pair	The same six pair observables, only in the two-tag category.
Three-lepton system	Invariant mass, transverse momentum, rapidity and scalar lepton H_{T}.
Visible system and event activity	Visible combinations use the three charged leptons and accepted tagged jets; jet activity counts all accepted jets. The catalogue includes visible mass and transverse momentum, scalar activity, S_{T} including missing momentum, missing transverse momentum, accepted-jet multiplicity and scalar jet H_{T}, and ordered lepton transverse momenta.

For objects a, b, pair quantities are evaluated on $p_a + p_b$; $\Delta R = \sqrt{(\eta_a - \eta_b)^2 + \Delta\phi^2}$. Four-vectors use (E, p_x, p_y, p_z), and $m = \sqrt{\max(0, p^2)}$. All energies and momenta are in GeV. The missing vector is calculated from all visible

status-one particles,

$$\vec{p}_{\mathrm{T}}^{\,\mathrm{miss}} = -\sum_{i \notin \{\nu_e, \nu_\mu, \nu_\tau\}} \vec{p}_{\mathrm{T},i},$$

including unaccepted radiation. Individual neutrino momenta are not used for any fiducial observable. The visible-system scalar sum and scalar jet sum have distinct titles so accepted light-jet activity is explicit.

5 Retained inclusive truth layer

The truth interpretation uses $e^+\nu_e$ for the top-side W^+, $e^-\bar{\nu}_e$ for the antitop-side W^-, and the muon and its charge-compatible neutrino for the associated W. Truth-only resonance, polarization and spin observables are filled only if this unique PDG content is present. The fiducial layer still fills for alternative parent-origin assignments and same-sign electrons.

Truth top momenta include decay radiation, identified by minimizing the sum of squared top/antitop mass residuals over the production/top/antitop assignment of each extra QCD parton. This uses exact neutrino momenta and is separate from the removed missing-momentum reconstruction. The retained lepton–bottom masses use recombined bottom jets rather than bare massless bottom partons. Merged bottom–antibottom jets do not fill tagged-jet or lepton–tag observables. The unresolvable assigned-parton-count histogram has been removed. Exact angular upper endpoints are placed in the final physical bin, including the Born back-to-back $\Delta\phi(W, t\bar{t}) = \pi$.

The truth spin basis is defined in the $t\bar{t}$ rest frame using the boosted positive beam $\hat{p}$ and top direction $\hat{k}$. With $c = \hat{p} \cdot \hat{k}$,

$$\hat{r} = \mathrm{sign}(c)\frac{\hat{p} - c\hat{k}}{\sqrt{1-c^2}}, \qquad \hat{n} = \mathrm{sign}(c)\frac{\hat{p} \times \hat{k}}{\sqrt{1-c^2}}.$$

The charged leptons in their parent rest frames supply six spin cosines, their nine products, and additional opening-angle and triple-product observables. Associated-W polarization and associated-lepton spin observables are retained at truth level only. The associated-W helicity axis is the negative $t\bar{t}$ recoil direction in the W rest frame; the charged-lepton analyser is boosted to that same frame.

6 Histogram numbering and normalization

IDs	Group
1–4	Inclusive rate, W charge, four-stage fiducial cutflow, fiducial W charge
5–34	Inclusive truth production
35–70	Inclusive truth charged leptons
71–94	Inclusive truth decay and polarization, with jet-based lepton–bottom masses
95–110	Truth neutrinos and missing momentum
111–138	Inclusive jet and event activity
139–164	Truth top-pair spin
165–170	Truth associated-lepton spin
171–173	Truth mass-assignment score, top mass residual, missing-momentum closure
174–309	Exclusive fiducial zero-tag category
310–445	Exclusive fiducial one-tag category
446–581	Exclusive fiducial two-tag category

The cutflow records valid object content, lepton acceptance, opposite-sign same-flavour mass threshold, and lepton–jet separation. Its last bin is the common fiducial baseline and has central value equal to the sum of the three exclusive category rates.

Histogram weights are signed fixed-order cross sections in pb per bin, not automatically divided by bin widths. All configured central, scale and PDF weights are filled. Object and pair histograms count their entries: when a species occurs twice, its integral can be twice the category rate. Only the category rate counts every accepted event exactly once. Truncated plot ranges can also limit histogram integrals. Bin errors include correlated subtraction contributions within an integration point; covariance between different bins must be retained separately for rebinned or normalized-shape error estimates.

The catalogue and fiducial definition changed on 15 October 2026. Old 286-histogram files contain the retired neutrino-fit observables and use different histogram numbering and cuts. They cannot be relabelled or recombined with the new histograms. Fresh integration output is required. On 19 October 2026 the lepton acceptance was

changed to the common $p_{\mathrm{T}} \geq 20$ GeV and $|\eta| < 2.5$ requirements and both Z-mass vetoes were removed. The cutflow now has four bins; the observable catalogue and its 581 histogram IDs are unchanged. Earlier output retains its original cuts and must not be combined with output using this selection.